Use Case: Tic Tac Toe Home Menu

Iteration: 1

Primary Actor: Player

Goal In Context: To enable the player to access the main game interface for Tic Tac Toe and select options such as starting a new game, viewing the leaderboard, or joining the matchmaking queue.

Preconditions:

- The player is logged in and has access to their profile.
- The Tic Tac Toe module is active.

Trigger: The player navigates to the Tic Tac Toe game section from the Player Profile Home Menu.

Scenario:

1. The player clicks “Tic Tac Toe” from the Player Profile Home Menu.
2. The system loads Tic Tac Toe Home Menu.
3. Player is given options to either view the Tic Tac Toe leaderboard or join the matchmaking queue.
4. The player selects their desired option (e.g., “Join Queue”).
5. The system directs the player to the chosen option.

Post Conditions: The player is successfully able to access their chosen feature in the Tic Tac module.

Exceptions:

- **E1:** The Tic Tac Toe module is unavailable. The system shows an error message and redirects the player to the Player Profile Home Menu.

Priority: High

When Available: When player is logged in & navigates to Tic Tac Toe

Frequency of Use: Frequent, as this is the starting point for Tic Tac Toe-related activities.

Channel to Actor: User Interface (UI)

Secondary Actors: None

Channels to Secondary Actors: N/A

Open Issues: N/A

Use Case: Tic Tac Toe Game Queue

Iteration: 1

Primary Actor: Player

Goal In Context: To allow the player to enter a matchmaking queue to find an opponent for the Tic Tac Toe game.

Preconditions:

- The player has accessed the Tic Tac Toe Home Menu.
- Matching making feature is working.

Trigger: The player selects the option to join the Tic Tac Toe game queue.

Scenario:

1. The player selects “Join Queue” from the Tic Tac Toe Home Menu.
2. The system adds the player to the Tic Tac Toe game queue.
3. The system searches for a opponent.
4. Once opponent is found, system notifies the player and initiates the Tic Tac Toe Game Instance.

Post Conditions: Player either joins a match or gets a message that a match is not currently available.

Exceptions:

- **E1:** No players are available in the queue, so the system displays a “No Match Found” message.
- **E2:** Matchmaking service is offline or is malfunctioning so the system shows an error message and returns the player to the Tic Tac Toe Home Menu.

Priority: High

When Available: When the player clicks to enter a queue.

Frequency of Use: Moderate, depending on matchmaking demand for users

Channel to Actor: UI notification

Secondary Actors: Matchmaking service

Channels to Secondary Actors: System-to-system communication with matchmaking service

Open Issues:

- Should players be matched based on certain skill level or expertise?
- Extension to above: If there isn't a player that can match skill levels, can the user be notified if & when a match is found?

Use Case: Tic Tac Toe Game Instance

Iteration: 1

Primary Actor: Player

Goal In Context: Enable the player to play a Tic Tac Toe game against another player.

Preconditions: A match has been successfully set up and an opponent is found from queue.

Trigger: The system begins the game instance when a suitable opponent is

Scenario:

1. System pairs player with it's opponent and initiates the game instance.
2. Player then receives a notification that the game is starting and is presented with the Tic Tac Toe game board
3. Then, the player and their opponent take turns making moves until the game ends (either a win, loss or draw)
4. The system records outcome of the game and updates the leaderboard with the respective winner.
5. The player is then directed to the result screen which displays "victory", "defeat" or "draw".
6. Player can choose to return to the Tic Tac Toe home screen or view the leaderboard from the results page.

Post Conditions: Game results are then recorded, and the player is directed to the game aftermath options.

Exceptions:

- **E1:** If the opponent disconnects or withdraws from game, player is notified and the game ends
- **E2:** If there is a connection issue on the player's end, system will save & freeze the game

Priority: High

When Available: Once matchmaking is done.

Frequency of Use: Varies, based on player's success in finding an opponent.

Channel to Actor: UI

Secondary Actors: Opponent user

Channels to Secondary Actors: Network communication for Tic Tac Toe moves

Open Issues:

- Should players be able to send challenges or game requests to recent opponents or opponents who are higher up on leaderboard?

Use Case: No Match Found

Iteration: 1

Primary Actor: Player

Goal In Context: Tells the player that the system was unable to find a match for them & it then provides options for them to exit the game or retry.

Preconditions: The player has joined the Tic Tac Toe Game Queue.

Trigger: If the system is unable to find an opponent for the player

Scenario:

1. The player joins the Tic Tac Toe Game Queue.
2. System searches for an opponent
3. After some time (5 minutes for ex), the system fails to find a match.
4. The system displays a "No Match Found" pop-up to the player

5. User is then given options to try joining the queue again or return to the Tic Tac Toe Home Menu.

Post Conditions: No match was found and player can choose to exit, retry later or try again now

Exceptions:

- **E1:** If there is an error getting the player's queue number, error message is shown & player is then redirected to Tic Tac Toe home menu

Priority: Medium

When Available: If the matchmaking queue times out or if there aren't enough players in the queue

Frequency of Use: Occasional

Channel to Actor: UI notification

Secondary Actors: None

Channels to Secondary Actors: N/A

Open Issues:

- Would implementing a "waiting room" feature with instructions for the game or mini challenges be a good way to increase user retention?

Use Case: Tic Tac Toe Leaderboard

Iteration: 1

Primary Actor: Player

Goal In Context: enables the player to view leaderboard for Tic Tac Toe & shows rankings in order of top player scores

Preconditions:

- The player passed the Tic Tac Toe Home Menu.

Trigger: The player selects the "Leaderboard" option from the Tic Tac Toe Home Menu.

Scenario:

1. The player navigates to the Tic Tac Toe Home Menu.
2. The player selects “Leaderboard.”
3. The system retrieves the latest leaderboard data for Tic Tac Toe.
4. The system displays the leaderboard, showing the following statistics for each player:
    - **Rank**: position based on win rate/score.
    - **Username**: display name of ranked players.
    - **Total Wins**
    - **Win Rate**: percentage of games won over total games.
    - **Longest Win Streak**: longest number of wins in a row.
    - **Games Played**: total number of games the player has done.
    - **Last Played Date**: date of most recent game.
    - **Moves to Win (Average)**: average number of moves player needs to take to win.
5. The player can scroll through or search the leaderboard to find specific players or rankings.
6. The player can filter the leaderboard (longest streak, most games played etc.)
7. The player can return to the Tic Tac Toe Home Menu

Post Conditions: The player has successfully viewed the Tic Tac Toe leaderboard and can navigate back to other Tic Tac Toe options.

Exceptions:

- **E1:** if the leaderboard data fails to load due to a network issue, the system will display an error message and provide the option to retry or return to the Tic Tac Toe Home Menu.
- **E2:** if the leaderboard is down, then the system informs the player and directs back to the Tic Tac Toe Home Menu.

Priority: Medium

When Available: Anytime the player is at the Tic Tac Toe Home Menu.

Frequency of Use: Occasional

Channel to Actor: UI display

Secondary Actors: Leaderboard database/service

Channels to Secondary Actors: System-to-database communication to retrieve leaderboard information

Open Issues:

- Would adding achievements, badges or premium features increase player retention and engagement?